

Product Requirements Document (PRD)

3D Forge — Custom 3D Products E-Commerce Platform

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1. Executive Summary

3D Forge is a B2C (with optional B2B) e-commerce platform that enables customers across India to browse and purchase pre-designed 3D products, as well as place orders for fully custom 3D-designed and 3D-printed items. The platform differentiates itself from conventional 3D printing businesses by being engineer-led, ensuring superior design quality, precision, and a premium product delivery experience.

The platform will be built on a modern stack — Next.js (frontend), Node.js + Express (backend), MongoDB (database), and Firebase (optional auth/storage) — and will support Razorpay payment integration with logistics partner integration for order tracking.

2. Problem Statement

The 3D printing and custom product market in India lacks platforms that:

- Allow customers to **visualize and order custom 3D products** online with ease
- Provide an **engineer-grade quality guarantee** rather than generic print services
- Offer a **seamless end-to-end experience** from product discovery → customization → checkout → delivery tracking

3D Forge addresses this gap by combining a curated product catalog with a robust custom order request flow, backed by a team of engineers.

3. Goals & Objectives

Business Goals

- Launch an MVP targeting youth and adult customers (ages 15–50) across India
- Establish brand credibility as an engineer-led 3D design & printing company
- Support both one-off B2C purchases and B2B bulk/custom orders

Product Goals

- Provide a frictionless product browsing and purchasing experience
- Enable customers to submit custom 3D product requests with specifications
- Give admins full control over product, order, and customer management
- Lay the groundwork for logistics partner API integration

Success Metrics (KPIs)

Metric	Target (6 months post-launch)
Monthly Active Users (MAU)	1,000+
Conversion Rate	≥ 2.5%
Average Order Value (AOV)	₹800–₹3,000
Custom Order Requests/month	100+
Cart Abandonment Rate	< 65%
Admin Order Processing Time	< 24 hrs

4. Target Audience

Primary — B2C

Segment	Description
Tech-savvy Youth (15–24)	Students, hobbyists, gamers, anime/pop culture fans wanting unique collectibles, phone stands, figurines
Young Professionals (25–35)	Working adults wanting personalized gifts, desk accessories, branded items
Adults & Gifting (35–50)	Buyers looking for custom gifting solutions, home décor, and novelty items

Secondary — B2B

- Small businesses needing branded merchandise or custom parts
- Event organizers needing trophies, nameplates, and promotional items
- Architects / interior designers needing scale models

Geography

- Phase 1: Pan-India (all tier 1 and tier 2 cities with delivery coverage)
- Phase 2: Expand based on logistics partner coverage

5. Scope

In Scope (MVP)

- Product catalog with categories and filters
- Custom product order request flow
- User authentication (email/password; OAuth optional)
- Shopping cart and checkout
- Razorpay payment integration (with fallback to manual/COD initially)
- Order management and tracking (manual + logistics API-ready)
- Admin dashboard
- Wishlist, reviews & ratings, coupons & discounts
- Search and filtering

Out of Scope (Post-MVP)

- Mobile app (Flutter)
- Multi-vendor/marketplace support
- AR/3D model preview in browser
- Subscription plans
- International shipping

6. User Roles & Permissions

Role	Description	Key Capabilities
Guest	Unauthenticated visitor	Browse products, view listings, search
Customer	Registered user	All guest actions + cart, checkout, wishlist, orders, reviews, custom requests
Admin	Platform operator	Full dashboard access: manage products, orders, users, coupons, analytics
Super Admin	Engineering team lead	All admin access + system settings, role management

7. Feature Requirements

7.1 User Authentication & Profiles

Description: Allow customers to register, log in, and manage their accounts.

Functional Requirements:

- Email + password registration with OTP email verification
- Login with email/password

- OAuth login (Google) — optional, Phase 2
- Forgot password / reset password flow
- User profile page: name, phone, address book (multiple addresses), order history
- Session management with JWT (access token + refresh token)
- Rate limiting on auth endpoints (prevent brute force)

Tech Notes:

- JWT stored in HTTP-only cookies (not localStorage)
- Firebase Auth as optional OAuth provider
- bcrypt for password hashing

7.2 Product Catalog & Listings

Description: A browsable, filterable catalog of pre-designed 3D products.

Functional Requirements:

- Product listing page with grid/list view toggle
- Each product card: image(s), name, price, rating, "Add to Cart" / "Add to Wishlist" buttons
- Product detail page:
 - Multiple product images (gallery with zoom)
 - 3D model preview (Phase 2 — Three.js viewer)
 - Full description, specifications (material, dimensions, weight, print time)
 - Price (with variant options if applicable: size, color, material)
 - Availability / stock status
 - Reviews & ratings section
 - Related products
- Product categories (e.g., Home Décor, Collectibles, Tech Accessories, Custom Parts, Gifts)
- Product tags for cross-category discoverability

Non-Functional Requirements:

- Images served via CDN (Cloudinary or Firebase Storage)
- Lazy loading for images
- SEO-optimized product URLs (/products/[slug])

7.3 Custom Product Order Request

Description: The core differentiator — allow customers to submit requests for fully custom 3D-designed and printed products.

Functional Requirements:

- Dedicated "Custom Order" page accessible from navbar and product pages
- Multi-step request form:
 - **Step 1 — Describe Your Idea:** Text description, reference images upload (up to 5 files, max 10MB each), preferred material, preferred color
 - **Step 2 — Dimensions & Quantity:** Approximate size (L×W×H in cm), quantity, deadline preference
 - **Step 3 — Contact & Budget:** Budget range (optional), preferred contact method (WhatsApp / Email), any additional notes
- Form submission creates a "Custom Order Request" record in DB with status: Pending Review
- Customer receives email confirmation with request ID
- Admin reviews and sends a **quote** back via email + platform notification
- Customer can accept/reject quote from their account dashboard
- Upon acceptance, a formal order is created and payment is triggered

Admin Workflow:

- View all custom order requests in admin dashboard
- Update request status: Pending Review → Quoted → Accepted → In Production → Shipped → Delivered
- Attach quote amount and estimated delivery date

7.4 Search & Filters

Description: Help users find products quickly.

Functional Requirements:

- Global search bar in navbar (real-time suggestions as user types)
- Search by product name, category, material, tags
- Filter sidebar on catalog page:
 - Category (multi-select)
 - Price range (slider)
 - Material (multi-select)
 - Rating (≥ 4 stars, etc.)

- Availability (In Stock only toggle)
- Sort options: Newest, Price Low→High, Price High→Low, Most Popular, Highest Rated
- Search results page with result count and active filter chips

Tech Notes:

- MongoDB text index for full-text search (Phase 1)
- Elasticsearch integration for advanced search (Phase 2)

7.5 Shopping Cart & Checkout

Description: Standard cart and checkout flow with Indian payment options.

Functional Requirements:

Cart:

- Add/remove items, update quantity
- Persistent cart (saved to DB for logged-in users, localStorage for guests — merged on login)
- Show item subtotal, quantity, and variant
- Apply coupon code at cart stage
- Show estimated shipping cost
- "Save for Later" (moves to wishlist)

Checkout (4-step flow):

1. **Address** — select saved address or add new; pin code-based state/city auto-fill
2. **Delivery** — estimated delivery date, shipping method (standard / express if applicable)
3. **Payment** — Razorpay (UPI, Cards, Net Banking, Wallets); COD option toggled by admin
4. **Confirmation** — Order summary, order ID, expected delivery

Non-Functional Requirements:

- Checkout must complete in < 3 seconds (excluding payment gateway)
- Handle payment failure gracefully with retry option
- Idempotent order creation (prevent duplicate orders on double-click)

7.6 Payment Gateway — Razorpay

Description: Integrate Razorpay as primary payment processor.

Functional Requirements:

- Razorpay Order creation via backend API
- Frontend Razorpay checkout modal integration

- Supported modes: UPI, Credit/Debit Card, Net Banking, Wallets
- Payment success webhook → update order status to Paid
- Payment failure → order stays Pending, user can retry
- Razorpay payment ID stored against each order for reconciliation
- COD (Cash on Delivery) as admin-configurable fallback option

Phase 1 Fallback:

- If Razorpay integration is delayed, support manual bank transfer with payment screenshot upload

Security:

- Razorpay signature verification on every webhook
- Payment amount validated on backend (never trust frontend-sent amount)

7.7 Order Management & Tracking

Description: Let customers track orders and give admins tools to manage fulfillment.

Functional Requirements:

Customer Side:

- "My Orders" page listing all orders with status badges
- Order detail page:
 - Items ordered, quantities, prices
 - Shipping address
 - Payment method & status
 - Order timeline / status tracker (visual stepper):
Order Placed → Payment Confirmed → In Production → Quality Check → Shipped → Delivered
 - Tracking number + logistics partner link (when available)
 - Cancel order option (if status is still Order Placed)
 - Download invoice (PDF)

Admin Side:

- View all orders, filter by status, date range, payment method
- Update order status manually
- Add tracking number and logistics partner name
- Logistics partner API integration (Shiprocket / Delhivery / EKart) — Phase 2
- Flag orders for issues / contact customer

Order Statuses:

Pending → Payment Confirmed → In Production → Quality Check → Shipped → Delivered → Cancelled / Refund Initiated / Refunded

7.8 Wishlist

Description: Let customers save products they're interested in.

Functional Requirements:

- Add/remove products from wishlist from catalog or product detail page
- Wishlist page in customer account
- "Move to Cart" button on wishlist items
- Wishlist is user-account-bound (not guest)
- Show if a wishlisted item's price has dropped (nice-to-have)

7.9 Reviews & Ratings

Description: Build trust through authentic customer reviews.

Functional Requirements:

- Star rating (1–5) + text review (optional)
- Only customers who have purchased the product can leave a review
- One review per customer per product (can edit)
- Reviews show: reviewer name (first name + last initial), date, rating, text, verified purchase badge
- Admin can hide/delete reviews that violate policy
- Average rating displayed on product card and detail page
- Sort reviews by: Most Recent, Highest Rated, Lowest Rated, Most Helpful

7.10 Coupons & Discounts

Description: Drive conversions and repeat purchases through promotional pricing.

Functional Requirements:

- Admin can create coupon codes with:
 - Code string (e.g., LAUNCH20)
 - Discount type: Flat amount (₹) or Percentage (%)
 - Maximum discount cap (for percentage coupons)
 - Minimum order value to apply

- Expiry date
- Usage limit (total / per user)
- Applicable to: all products / specific categories / specific products
- Customers enter code at cart stage
- Real-time validation with error messages (expired, invalid, minimum not met, already used)
- Admin dashboard shows coupon usage analytics

7.11 Admin Dashboard

Description: Full control panel for the engineering/operations team.

Sections:

Section	Key Features
Overview	KPI cards: Total Revenue, Orders Today, Pending Orders, New Users, Conversion Rate
Products	Add / Edit / Delete products; upload images; manage stock; set variants
Custom Orders	View requests, update status, send quotes, accept/reject
Orders	List all orders, filter/sort, update status, add tracking info
Users	View customer list, search, view order history, flag/block accounts
Coupons	Create/edit/deactivate coupons, view usage
Reviews	Moderate reviews, hide/delete
Analytics	Revenue charts, top products, traffic sources (Phase 2)
Settings	Site settings, payment modes, logistics toggle, shipping rates

Tech Notes:

- Admin routes protected by role-based middleware on backend
- Admin UI built with Next.js (separate /admin route group with layout)

8. Non-Functional Requirements

8.1 Performance

- Page load time (LCP): < 2.5 seconds on 4G connection
- API response time (p95): < 500ms
- Image optimization: WebP format, CDN-served, lazy loaded

8.2 Security

- HTTPS enforced across all routes
- JWT in HTTP-only cookies
- Input sanitization and validation on all API endpoints (Joi / Zod)
- Rate limiting on auth, checkout, and coupon endpoints
- Razorpay webhook signature verification
- No sensitive data (passwords, card numbers) stored in DB

8.3 Scalability

- MongoDB Atlas with connection pooling
- Stateless Express API (ready for horizontal scaling)
- CDN for static assets and product images

8.4 Availability

- Target uptime: 99.5% (Phase 1)
- MongoDB Atlas auto-backups enabled
- Error monitoring: Sentry integration

8.5 SEO & Discoverability

- Next.js SSR/SSG for product pages
- OpenGraph meta tags for social sharing
- Structured data (JSON-LD) for products (Google Shopping compatibility)
- Sitemap auto-generation

8.6 Accessibility

- WCAG 2.1 AA compliance
- Keyboard navigable checkout and forms
- Proper alt text on all product images

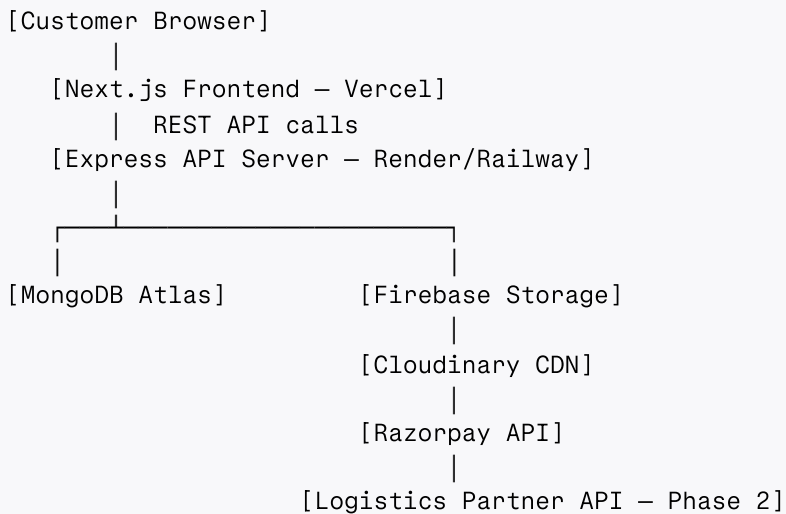
9. Technical Architecture

Tech Stack

Layer	Technology
Frontend	Next.js 14+ (App Router), Tailwind CSS, shadcn/ui
Backend	Node.js, Express.js (REST API)

Layer	Technology
Database	MongoDB Atlas (Mongoose ODM)
Auth	JWT (primary), Firebase Auth (OAuth — optional)
File Storage	Firebase Storage / Cloudinary (product images, custom order files)
Payments	Razorpay
Email	Nodemailer + Gmail SMTP / Resend
Hosting	Vercel (Frontend), Render / Railway (Backend)
CDN	Vercel Edge / Cloudinary
Monitoring	Sentry

High-Level Architecture



Key Data Models (MongoDB)

User: `_id`, `name`, `email`, `passwordHash`, `role`, `addresses[]`, `createdAt`

Product: `_id`, `title`, `slug`, `description`, `category`, `tags[]`, `images[]`, `variants[]`
(`size/color/material + price + stock`), `averageRating`, `reviewCount`, `isActive`

CustomOrderRequest: `_id`, `userId`, `description`, `referenceImages[]`, `material`, `dimensions`,
`quantity`, `budgetRange`, `status`, `quoteAmount`, `adminNotes`, `createdAt`

Order: `_id`, `userId`, `items[]`, `shippingAddress`, `paymentMethod`, `paymentStatus`, `razorpayOrderId`,
`razorpayPaymentId`, `couponApplied`, `subtotal`, `discount`, `shippingCost`, `total`, `status`,
`trackingNumber`, `logisticsPartner`, `timeline[]`, `createdAt`

Coupon: `_id`, `code`, `discountType`, `discountValue`, `maxDiscount`, `minOrderValue`, `expiresAt`,
`usageLimit`, `usedCount`, `applicableTo`

Review: `_id, productId, userId, orderId, rating, comment, isVerifiedPurchase, isHidden, createdAt`

10. User Flows

10.1 Standard Purchase Flow

```
Landing Page
  → Browse Catalog (filter/search)
    → Product Detail Page
      → Add to Cart
        → Cart Review + Apply Coupon
          → Login / Guest Checkout
            → Enter Address
              → Select Payment Method
                → Razorpay Payment
                  → Order Confirmation Page
                    → Email Confirmation
```

10.2 Custom Order Flow

```
"Request Custom Product" CTA
  → Login Required
    → Custom Order Form (3 steps)
      → Submit Request
        → Admin Reviews → Sends Quote
          → Customer Notified (Email + In-app)
            → Customer Accepts Quote
              → Payment Link Generated
                → Order Created → Production → Delivery
```

11. Phased Rollout

Phase 1 — MVP (Month 1–3)

- [x] Product catalog (static listings)
- [x] User auth (email/password)
- [x] Cart + checkout
- [x] Razorpay integration
- [x] Custom order request form
- [x] Order tracking (manual status updates)
- [x] Admin dashboard (core: products, orders, users)
- [x] Wishlist

- ☒ Reviews & ratings
- ☒ Basic coupon system
- ☒ Search + filters
- ☒ Email notifications (order confirmation, status updates)

Phase 2 — Growth (Month 4–6)

- ☐ OAuth (Google login via Firebase)
- ☐ Logistics partner API integration (Shiprocket/Delhivery)
- ☐ Advanced admin analytics
- ☐ 3D model preview on product page (Three.js)
- ☐ Push notifications
- ☐ B2B bulk order flow
- ☐ Flutter mobile app

Phase 3 — Scale (Month 7+)

- ☐ AR product preview
- ☐ Elasticsearch for advanced search
- ☐ Affiliate/referral program
- ☐ Multi-city express delivery options
- ☐ Loyalty points system

12. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Razorpay integration delay	Medium	High	Build COD / manual transfer fallback first
Logistics partner API not ready	High	Medium	Start with manual tracking + notify customers via email
Custom order scope creep	Medium	Medium	Strict form fields; free-form communication via email after submission
Image storage costs	Low	Low	Compress images client-side before upload; use free Cloudinary tier first
MongoDB query performance	Low	Medium	Add indexes on slug, category, userId, status fields from day 1
Admin overwhelmed with requests	Medium	Medium	Set clear SLA (respond to custom requests within 48 hrs) in UI

13. Open Questions

1. **Shipping rates:** Will shipping be free, flat-rate, or weight/zone-based? This affects checkout UI complexity.
2. **COD availability:** Which pin codes / states will COD be enabled for?
3. **Custom order turnaround:** What's the typical production time? Should this be communicated upfront per product?
4. **Returns/refund policy:** For 3D printed custom items, what's the return window and policy? (Custom items are typically non-returnable.)
5. **B2B pricing:** Will B2B customers get different pricing tiers or bulk discounts?
6. **Brand name:** Is "3D Forge" the confirmed brand name, or is this TBD?
7. **Product variants:** Do products have multiple material/color variants with separate pricing, or is pricing fixed per product?

14. Appendix — API Endpoint Overview (Express)

Auth

- POST /api/auth/register
- POST /api/auth/login
- POST /api/auth/logout
- POST /api/auth/refresh
- POST /api/auth/forgot-password
- POST /api/auth/reset-password

Products

- GET /api/products (with query params for filter/sort/paginate)
- GET /api/products/:slug
- POST /api/products (*Admin*)
- PUT /api/products/:id (*Admin*)
- DELETE /api/products/:id (*Admin*)

Custom Orders

- POST /api/custom-orders (*Customer*)
- GET /api/custom-orders (*Admin*)
- GET /api/custom-orders/:id
- PUT /api/custom-orders/:id/status (*Admin*)
- PUT /api/custom-orders/:id/accept-quote (*Customer*)

Cart

- GET /api/cart
- POST /api/cart/add
- PUT /api/cart/update
- DELETE /api/cart/remove

Orders

- POST /api/orders (create order)
- GET /api/orders (*Admin — all orders*)
- GET /api/orders/my (*Customer — own orders*)
- GET /api/orders/:id
- PUT /api/orders/:id/status (*Admin*)
- POST /api/orders/:id/cancel (*Customer*)

Payments

- POST /api/payments/create-order (Razorpay order creation)
- POST /api/payments/verify (signature verification)
- POST /api/payments/webhook (Razorpay webhook)

Reviews

- POST /api/reviews
- GET /api/reviews/product/:productId
- PUT /api/reviews/:id
- DELETE /api/reviews/:id (*Admin*)

Wishlist

- GET /api/wishlist
- POST /api/wishlist/add
- DELETE /api/wishlist/remove/:productId

Coupons

- POST /api/coupons/validate
- POST /api/coupons (*Admin*)
- GET /api/coupons (*Admin*)
- PUT /api/coupons/:id (*Admin*)

